

Why the loop works: a learned path to the clean manifold, held on the physics at every step

the prior supplies the DIRECTION (it was trained on this very path); the measured data supplies the CONSTRAINT that keeps the direction honest

ONE STEP OF THE LOOP (x50)

-  **PREDICT** — the prior steps toward $\mathcal{M}_{clean}$
it knows the direction: it was TRAINED on this path
-  **ESTIMATE** — at the vertex between the two arrows:
 $\hat{\theta}$ is refit on the image the prior just made.
It moves θ , not x — so $c(\hat{\theta})$ (dashed) re-aims instead.
-  **CORRECT** — the data step pulls it back onto $c(\hat{\theta})$; the TV corrector keeps it regular

prior alone:
a plausible image that does NOT
explain the measurement

$\mathcal{M}_{clean}$
motion-free images —
what the scan would have been

landing: $\tilde{\theta}$ and the carried x_t

the ESTIMATE stage is what re-aims it: $c(\hat{\theta}_0) \rightarrow c(\hat{\theta}_k) \rightarrow c(\theta_{true})$,
until the feasible set finally CONTAINS the clean image.

$c(\hat{\theta}) = \{x : A_{\hat{\theta}}x \approx y\}$ — the constraint that PHYSICALLY EXISTS:
the measured projections. It is not a regularizer we chose, it is the scan.

START

$$x_0 = \text{FDK}(y, P_{nom})$$

streaked — and consistent
with the WRONG geometry

$c(\hat{\theta}_0)$

GEOMETRY BRIDGE $\{\text{FDK}(y, P_{nom}T(s\theta))\}$

every point on it is a REAL reconstruction of the REAL data —
and it is exactly the path the prior was trained on ($v_{\psi} \approx dx_t/dt$)

data step alone, at the geometry we started with ($\hat{\theta} = 0$): perfectly consistent — and still streaked

the three moves alternate $N = 50$ times — no single one reaches $\mathcal{M}_{clean}$